

Data Science (CSE558)

Mid-Sem Solutions

1. There is a fair coin, a 4-faced die, and an 8-faced die. Answer the following questions:

- a) Let the coin and both dice are unbiased. Now, first the fair coin is tossed. If it lands heads, you roll a fair 4-faced die and receive candies equal to twice the number shown. If it lands tails, you roll a fair 8-faced die and lose candies equal to the number shown. What is the expected number of candies you will have after one play of the game? **(3 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: First found the expected candies for each outcome of the coin toss (Heads or Tails) and then compute their weighted average based on the 50% probability of each outcome.

Ans: Let X be the random variable for the number of candies.

1. **If Heads (H):** Roll a 4-faced die (D_4) and get $2 \times \text{value}$.

$$\circ E[D_4] = \frac{1+2+3+4}{4} = 2.5$$

$$\circ \text{Expected candies if Heads: } 2 \times E[D_4] = 2 \times 2.5 = 5.$$

2. **If Tails (T):** Roll an 8-faced die (D_8) and lose candies equal to the value.

$$\circ E[D_8] = \frac{1+2+3+4+5+6+7+8}{8} = 4.5.$$

$$\circ \text{Expected candies if Tails: } -E[D_8] = -4.5$$

3. As the coin is unbiased, $P(H)=P(T)=0.5$.

$$\circ \text{Total Expectation: } E[D_4] \times 0.5 - E[D_8] \times 0.5 = 0.25$$

The expected number of candies is 0.25.

Rubric:

1. Full marks for correct calculations, i.e., total expected candies = 0.25.
2. For incorrect total expectation but correct individual expected candies calculation award 2 marks (1 mark each for $E[D_4]$ and $E[D_8]$).

- b) You and your friend are playing a game using the coin and both dice. You both toss the coin simultaneously until one of the following sequences appears: HT or TT. If HT occurs, you roll the 8-faced die and receive candies equal to the face value. If TT occurs, your friend rolls the 4-faced die and receives candies equal to the face value. What is the expected number of candies you will receive from this game? **(4 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: The game ends with either HT or TT. Your friend cannot win if first toss is H, making your probability of winning 75%. Your total expected winnings are found by multiplying the expected candies you get upon winning (the average of an 8-sided die) by your probability of winning.

Ans: Probability of Winning: We consider that the game always results into either you win or your friend wins. Further, notice that if the first toss is heads then you friend can never win. Hence, $P(\text{You win}) = P(\text{HT comes before TT}) = \frac{3}{4}$. Now, the expected number of candies from D_8 is $E[D_8] = \frac{36}{8} = 4.5$. Hence, expected number of candies that you can win is $\frac{36}{8} \times \frac{3}{4} = \frac{27}{16}$.

Rubric

1. Full marks for correct answer with calculations.
2. Deduct two marks if $P(\text{You win}) = P(\text{HT comes before TT})$ is not equal to $\frac{3}{4}$. Award 0.5 marks if $P(\text{You win}) = P(\text{HT comes before TT}) = 0.5$
2. A national coding challenge will award 150 internship positions to the top 150 participants based on their scores in a problem-solving assessment. The total number of participants who took the test is unknown. The problem-solving scores are nonnegative, with a mean of 40 and a standard deviation of 8. Your friend, Ravi, scored 85 on the test, and his score is unique — no other participant has scored exactly 85. The challenge rules state that the top 150 scores will be selected for the internships. What is the minimum number of participants who must have taken the test for Ravi to expect getting an internship positions? **(4 Marks)**
Ignore this question.
3. We roll a standard fair 6-faced die over and over, where rolls are independent of each other. What is the expected number of rolls until the first pair of consecutive sixes appears? **(5 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Define two states: one where the previous roll was not a six, and one where it was. By creating equations for the expected number of future rolls from each state using conditional probability, solve the resulting system of equations to find the initial expected value.

Ans: Let E be the expected number of rolls.

- **State 0 (E_0):** The initial state (last roll was not a six). $E = E_0$.
 - **State 1 (E_1):** The state after rolling one six
1. **Equation from State 0:** A roll costs 1 try.
 - With probability $1/6$, we roll a six and go to State 1.
 - With probability $5/6$, we roll a non-six and return to State 0.
 - $E_0 = 1 + \frac{1}{6}E_1 + \frac{5}{6}E_0 \rightarrow E_0 = 6 + E_1$.
 2. **Equation from State 1:** A roll costs 1 try.
 - With probability $1/6$, we roll another six and the game ends (0 more rolls needed).
 - With probability $5/6$, we roll a non-six and go back to State 0.
 - $E_1 = 1 + \frac{5}{6}E_0$.
 3. **Solve the System:** we have: $E_0 = 42$. The expected number of rolls is 42.

Rubric

1. Full marks for correct answer and correct analysis.
 2. Deduct 1 if the final answer is not 42.
 3. Deduct 4 marks if incorrect formulation of the expected number of required toss.
 4. Award 1.5 marks mark for correctly stating state similar to E_0
 5. Award 1.5 marks mark for correctly stating state similar to E_1
4. A standard die is rolled 120 times, and the results are: 1 (20 times), 2 (22 times), 3 (17 times), 4 (18 times), 5 (19 times), 6 (24 times). Perform a test to determine if the die is fair at a 5% significance level. **(5 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Use the Chi-squared goodness-of-fit test by comparing the observed frequencies of die rolls to the expected frequencies for a fair die. The calculated Chi-squared statistic was smaller than the critical value for a 5% significance level. Thus, the null hypothesis that the die is fair cannot be rejected.

Ans: Chi square goodness of fit test.

H0: The die is unbiased

H1: The die is biased.

- Compute expected outcomes, i.e., $\frac{1}{6}$ of 120 tosses for each face. 1(20 times), 2 (20 times), so on.
- Compute test statistic (refer formula sheet) for chi square. It will be $\frac{34}{20}$.
- Here, degree of freedom is $6-1 = 5$.
- Hence, for 5% level of significance the critical points are 1.15 and 11.07. Since test statistics is between this range hence we do not reject null hypothesis (H0).

Rubric

1. Award 1 mark for stating the correct null and alternative hypotheses.
 2. Award 1 mark for correctly calculating the expected frequencies (20 for each).
 3. Award 1.5 marks for correctly calculating the χ^2 statistic (1.7).
 4. Award 1.5 marks for identifying the correct degrees of freedom (5), critical value (11.07), and making the correct final conclusion.
5. Suppose there is a large collection of coupons of two distinct types, Type I and Type II. The number of Type I coupons is exactly double the number of Type II coupons. If you draw coupons one at a time at random and then return them to the collection, what is the expected number of draws required to collect at least one of each type? **(5 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: After the first draw, the problem becomes waiting for a specific coupon type, which follows a geometric distribution. calculate the total expected draws for both possible first-draw scenarios (Type I or Type II) and then computed the weighted average of these outcomes based on their initial probabilities.

Ans:

- **Probabilities:** Let p_1 be the probability of drawing Type I and p_2 be for Type II.
 - o Since Type I is double Type II, $p_1=2/3$ and $p_2=1/3$.
- **Expected Draws:** After the first draw, we need to wait for the other type. This waiting time follows a geometric distribution.
- **Case 1: First draw is Type I.** (This happens with probability $2/3$, i.e., $p_1=2/3$).
 - o Now, we need a Type II. The number of *additional* draws needed is $1/p_2=3$.
 - o Total expected draws in this case: $1+3=4$
- **Case 2: First draw is Type II.** (This happens with probability $p_2=1/3$).
 - o Now we need a Type I. The number of *additional* draws needed is $1/p_1=1.5$.
 - o Total expected draws in this case: $1+1.5=2.5$.
- **Overall Expectation:**
 - o $E[\text{Draws}] = (\text{Prob. Case 1} \times \text{Exp. Draws Case 1}) + (\text{Prob. Case 2} \times \text{Exp. Draws Case 2})$
 - o $E[\text{Draws}] = (2/3 \times 4) + (1/3 \times 2.5) = 3.33 + 0.83 = 4.17$

Rubric

1. Award 1 mark for correctly identifying the probabilities of drawing each coupon type ($2/3$ and $1/3$).
2. Award 2 marks for correctly calculating the expected total draws for both cases (4 and 2.5).
3. Award 2 marks for correctly applying the law of total expectation to find the final answer (3.5).

6. At an airport, only 2% of checked bags contain prohibited items. The security scanner correctly flags bags with prohibited items 98% of the time, and it incorrectly flags clean bags 2% of the time. Suppose your friend asked you to bring a medicine for him.
- a) If your bag is flagged by the scanner, probabilistically, how concerned should you be that it actually contains a prohibited drug? **(4 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: In this case, I will be concerned if my bag actually contains prohibited items in the medicine given that it has been flagged.

Ans: Let D be the event the bag has a prohibited item, and F be the event the scanner flags it.

Given:

- $P(D) = 0.02 \Rightarrow P(D^c) = 0.98$
- $P(F|D) = 0.98$ (True Positive Rate)
- $P(F|D^c) = 0.02$ (False Positive Rate)

Now, in this case, I will be concerned if my bag actually contains prohibited items in the medicine given that it has been flagged, i.e., $P(D|F)$.

$$P(D|F) = \frac{P(F|D) \times P(D)}{P(F)} = 0.5$$

Hence, this is a highly concerning probability.

Rubric:

1. Award 1 mark for correctly setting up the problem with given probabilities.
2. Award 1.5 marks for correctly calculating the total probability of a flag, $P(F)$.
3. Award 1.5 marks for correctly applying Bayes' theorem to get the final answer (0.5)

- b) You go to a second, independent scanner at another checkpoint, and this time your bag is not flagged. Probabilistically, how relieved should you be now? **(6 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Now, I update my prior, i.e., $P(D)$. Rest everything remains same. Here, I will be relieved if $P(D|F^c)$ is low or $P(D^c|F^c)$ is high.

Ans: Update prior,

New Prior: Our new belief after the first flag is $P(D) = P(D|F) = 0.5$. So $P(D^c) = 0.5$.

Compute: $P(F^c|D) = 1 - P(F|D) = 0.02$ and $P(F^c|D^c) = 1 - P(F|D^c) = 0.98$.

Rest remains the same.

Now, in this case, I will be relieved if my bag does not contains prohibited items in the medicine given that it has not been flagged, i.e., $P(D|F^c)$.

$$P(D|F^c) = \frac{P(F^c|D) \times P(D)}{P(F^c)} = 0.02$$

This is a reliving probability.

Rubric

1. Award 1 mark for identifying the posterior from part (a) as the new prior.
2. Award 2 marks for correctly identifying the likelihoods for the two events $P(F^c|D)$ and $P(F^c|D^c)$.
3. Award 3 marks for correctly applying the second Bayesian update to get the final answer. Note, that student can compute $P(D|F^c)$ or $P(D^c|F^c)$. Either of them are right. They must calculate $P(D|F^c) = 0.02$ or $P(D^c|F^c) = 0.98$.

7. You are handling a project for a university library system. To automate monitoring of book circulation, you use Bloom filters with the same k hash functions
- BF_borrowed: Each time a student borrows a book, their ID is hashed into this Bloom filter.
 - BF_returned: Each time a student returns a book, their ID is hashed into this Bloom filter.
- A student can at max borrow or return one book. Now design an algorithm with two different functions that answers the following questions. **(7 Marks)**
- a) Count approximate number of sincere students who have borrowed and returned a book.

Evaluate if all or one the explanation points is mentioned.

Explanation: If a student has borrowed and returned a book then the student must be present in both the bloom filters, i.e., BF_borrowed and BF_returned. For every student, check this fact and increment a counter whenever its true.

Ans:

1. count = 0
2. Compute BF = biwiseAND(BF_borrowed, BF_returned)
3. For every student s in the university
 1. If query(s ,BF) = True then
 - a. count = count + 1
4. Return count

Rubric

1. The idea is to use set intersection of both bloom filters, i.e., $\text{BF_borrowed} \cap \text{BF_returned}$.

- b) Count approximate number of students who have borrowed a book but did not return yet.

Evaluate if all or one the explanation points is mentioned.

Explanation: If a student has borrowed and but not returned a book then the student must be present in BF_borrowed and not present in BF_returned. For every student, check this fact and increment a counter whenever its true.

Ans:

1. countBR = 0; countB = 0;
2. Compute BF = biwiseAND(BF_borrowed, BF_returned)
3. For every student s in the university
 1. If query(s ,BF) = True then
 - a. countBR = countBR + 1
 2. If query(s ,BF) = True then
 - a. countB = countB + 1
4. Return countB – countBR.

Rubric

1. The idea is to use set difference, i.e., $\text{BF_borrowed} \setminus \text{BF_Returned}$.

8. A university wellness office surveys students about their exercise frequency and stress. Rows represent exercise frequency; columns represent self-reported stress.

Observed	Not Stressed	Stressed	Total
Rarely exercise	10	30	40
Sometimes exercise	18	22	40

Often exercise	17	3	20
Total	45	55	100

- a) It is assumed that at most 50% of students are stressed irrespective of exercise habits. With a 5% level of significance, what can you conclude from the above table? **(3 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Perform a one-sided Z-test for a population proportion, comparing the observed sample proportion of stressed students (55%) against the hypothesized proportion (50%). The calculated Z-score was not extreme enough to fall into the rejection region, so we cannot reject the initial assumption.

Ans

This is a one-sided z-test. Now, define null hypothesis H_0 and alternate hypothesis H_1 .
 $H_0: p \leq 0.50$ (The proportion of stressed students is at most 50%).

$H_1: p > 0.50$ (The proportion is greater than 50%). This is a right-tailed test.

Here, $p = 0.55$ and $p_0 = 0.5$. Now computing z-statistics we get, 1.0.

At 5% level of significance, since critical region starts at 1.645. Hence we do not have enough evidence to reject null hypothesis (H_0).

Rubric:

1. Award 0.5 marks for correctly identifying the test.
2. Award 1 mark if null hypothesis or alternate hypothesis are incorrectly stated.
3. Award 0.5 mark for correctly using the value of p or p_0 in the z-test formula.
4. Award 1 mark if final conclusion is correct.

- b) Is there enough evidence in the data to suggest that, at the 5% level of significance, exercise frequency changes student's stress status? **(7 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Use a Chi-squared test of independence to assess the relationship between exercise and stress. Compare the observed data against expected frequencies calculated under the assumption of independence. The resulting Chi-squared value was significantly larger than the critical threshold, leading to the rejection of independence and the conclusion that the two variables are associated.

Ans

This is a Chi-squared (χ^2) test of independence. Define H_0 and H_1 .

H_0 : Exercise frequency and stress status are independent.

H_1 : Exercise frequency and stress status are dependent.

The *Expected* = $\frac{\text{Row Sum} \times \text{Column Sum}}{\text{Total}}$.

Now, compute the expected values.

Expected	Not Stressed	Stressed
Rarely	$(40 \times 45)/100 = 18$	$(40 \times 55)/100 = 22$
Sometimes	$(40 \times 45)/100 = 18$	$(40 \times 55)/100 = 22$
Often	$(20 \times 45)/100 = 9$	$(20 \times 55)/100 = 11$

Calculating χ^2 returns 19.34.

Here, the degree of freedom is $(\text{\#rows} - 1) \times (\text{\#columns} - 1) = 2$.

At 5% level of significance for 2 degree of freedom the critical points are 0.1 and 5.99.
Hence, we have enough evidence to reject the null hypothesis.

Rubric:

1. Award 1 marks for correctly identifying the test.
2. Award 1 mark if null hypothesis or alternate hypothesis are incorrectly stated.
3. Award 3 marks for correctly calculating the expected values.
4. Award 1 mark if degree of freedom is correctly mentioned.
5. Award 1 mark if final conclusion is correct.

9. **(Bonus)** You manage quality assurance for a pharmaceutical plant. Each batch of a liquid formulation must have an average active ingredient concentration between 2 and 5 mg/mL by design. Individual vial measurements, denoted, $X_1, X_2, \dots, X_n$ are i.i.d. and bounded in $[a, b]$ mg/mL. You want to certify a batch by sampling vials and estimating the batch mean 3 with the sample mean $\bar{x}$. Regulatory policy requires that the probability your estimate is not off by more than $\epsilon = 0.05$ mg/mL is at least 0.99. Determine the smallest sample size n that guarantees this requirement. Clearly show the bound you use and the final numerical value of n . **(7 Marks)**

Evaluate if all or one the explanation points is mentioned.

Explanation: Use Hoeffding's inequality, which bounds the probability of a sample mean deviating from the true mean for bounded data. By setting this probability bound equal to the maximum acceptable failure rate ($1 - 0.99 = 0.01$) and using the given error tolerance and data range, derive an inequality and solved it for the minimum sample size n .

Ans:

We want the probability that the sample mean $\bar{X}$ is off by more than $\epsilon = 0.05$ to be at most 0.01.

$$P(|\bar{X} - \mu| \geq 0.05) \leq 0.01.$$

From Hoeffding we know,

$$P(|\bar{X} - \mu| \geq \epsilon) \leq 2 \exp \left(\frac{-2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)} \right)$$

Since $\bar{X}$ is the average of all the n samples, hence the each random variables i is defined as $\frac{X_i}{n}$. Hence, their range is $\left[\frac{2}{n}, \frac{5}{n} \right]$. Substituting, values in the above expression and bounding it with 0.01 we get, $n \geq 9537$.

Rubric:

1. Award 1.5 marks for correctly identifying Hoeffding's Inequality as the correct tool.
2. Award 1 mark for correctly identifying the range of each random variable.
3. Award 1 mark for correctly setting up the inequality with all parameters (ϵ , $b-a$, probability bound) correctly substituted.
4. Award 2.5 marks for correctly showing the algebraic steps to solve for n , including the use of logarithms.
5. Award 1 mark for arriving at the correct final integer answer.

Formula sheet

z-tests (known variance & proportional)

$$Z = \frac{p - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

TABLES

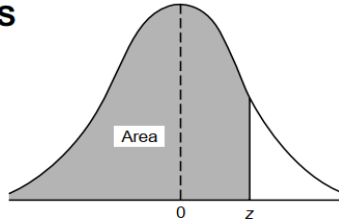


Table 1 The normal curve

(a) Area under the normal curve

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0017	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0124	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0352	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0722	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379

$$Z_{0.05} = -1.645$$

$$Z_{0.0018} = -2.91$$

- $-2.91 = -(2.9 + 0.01)$, i.e, In the above table, the first column with value -2.9 and first row with 0.01.
- So, at 0.18% level of significance, the left curve critical value is -2.91.
- Since Z-scores follows a normal distribution, so you may use its property of symmetry.

Markov's Inequality: Let X be a random variable that takes non negative values. Then for all $a > 0$,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's Inequality: Let X be a random variable. Then for any $a > 0$,

$$P(|X - E[X]| \geq a) \leq \frac{\text{Var}[X]}{a^2}$$

Hoeffding's Inequality: Let $X_1, X_2, \dots, X_n$ be n independent random variable such that $a_i \leq X_i \leq b_i$. Let $S = X_1 + X_2 + \dots + X_n$. Then for all $t > 0$,

$$P(|S - E[S]| \geq t) \leq 2 \cdot \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

χ^2 -test

$$\chi^2 = \sum_{i=1}^K \frac{(O_i - E_i)^2}{E_i}$$

v: Degree of freedom

a: Left critical value.

b: Right critical value.

	Level of significance α							
	0.20		0.10		0.05		0.01	
Two-sided	0.10		0.05		0.025		0.005	
One-sided								
ν	a	b	a	b	a	b	a	b
1	0.016	2.71	39.10 ⁻⁴	3.84	98.10 ⁻⁵	5.02	16.10 ⁻⁵	6.63
2	0.21	4.61	0.10	5.99	0.05	7.38	0.02	9.21
3	0.58	6.25	0.35	7.81	0.22	9.35	0.11	11.34
4	1.06	7.78	0.71	9.49	0.48	11.14	0.30	13.28
5	1.61	9.24	1.15	11.07	0.83	12.83	0.55	15.09

t-test

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

$$s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n - 1}$$

Degrees of freedom	Two-tailed test: One-tailed test:	Significance level					
		10% 5%	5% 2.5%	2% 1%	1% 0.5%	0.2% 0.1%	0.1% 0.05%
1		6.314	12.706	31.821	63.657	318.309	636.619
2		2.920	4.303	6.965	9.925	22.327	31.599
3		2.353	3.182	4.541	5.841	10.215	12.924
4		2.132	2.776	3.747	4.604	7.173	8.610
5		2.015	2.571	3.365	4.032	5.893	6.869
6		1.943	2.447	3.143	3.707	5.208	5.959
7		1.894	2.365	2.998	3.499	4.785	5.408
8		1.860	2.306	2.896	3.355	4.501	5.041
9		1.833	2.262	2.821	3.250	4.297	4.781
10		1.812	2.228	2.764	3.169	4.144	4.587
11		1.796	2.201	2.718	3.106	4.025	4.437
12		1.782	2.179	2.681	3.055	3.930	4.318
13		1.771	2.160	2.650	3.012	3.852	4.221
14		1.761	2.145	2.624	2.977	3.787	4.140
15		1.753	2.131	2.602	2.947	3.733	4.073
16		1.746	2.120	2.583	2.921	3.686	4.015
17		1.740	2.110	2.567	2.898	3.646	3.965
18		1.734	2.101	2.552	2.878	3.610	3.922
19		1.729	2.093	2.539	2.861	3.579	3.883
20		1.725	2.086	2.528	2.845	3.552	3.850
21		1.721	2.080	2.518	2.831	3.527	3.819
22		1.717	2.074	2.508	2.819	3.505	3.792
23		1.714	2.069	2.500	2.807	3.485	3.768
24		1.711	2.064	2.492	2.797	3.467	3.745
25		1.708	2.060	2.485	2.787	3.450	3.725
26		1.706	2.056	2.479	2.779	3.435	3.707
27		1.703	2.052	2.473	2.771	3.421	3.690
28		1.701	2.048	2.467	2.763	3.408	3.674
29		1.699	2.045	2.462	2.756	3.396	3.659
30		1.697	2.042	2.457	2.750	3.385	3.646
32		1.694	2.037	2.449	2.738	3.365	3.622
34		1.691	2.032	2.441	2.728	3.348	3.601
36		1.688	2.028	2.434	2.719	3.333	3.582
38		1.686	2.024	2.429	2.712	3.319	3.566
40		1.684	2.021	2.423	2.704	3.307	3.551
42		1.682	2.018	2.418	2.698	3.296	3.538
44		1.680	2.015	2.414	2.692	3.286	3.526
46		1.679	2.013	2.410	2.687	3.277	3.515
48		1.677	2.011	2.407	2.682	3.269	3.505
50		1.676	2.009	2.403	2.678	3.261	3.496
60		1.671	2.000	2.390	2.660	3.232	3.460
70		1.667	1.994	2.381	2.648	3.211	3.435
80		1.664	1.990	2.374	2.639	3.195	3.416
90		1.662	1.987	2.368	2.632	3.183	3.402
100		1.660	1.984	2.364	2.626	3.174	3.390